National University of Computer and Emerging Sciences

Karachi Campus

CS-303 Software Engineering (Midterm I Exam) - Spring 2012

Dated: 24 Feb 2012 Time Duration: 90 mins

ATTEMPT ALL QUESTIONS; TOTAL MARKS: 50

SECTION A (10 MARKS; 1 MARKS EACH. In order to get a correct answer, both the Choice and a single sentence reason / example is required.)

1. Which of the items listed below is not one of the software engineering layers?

   a. Process
   b. Manufacturing
   c. Methods
   d. Tools

   Why: _____

2. Which of these terms is a level name in the Capability Maturity Model?

   a. Inexistent
   b. Defined
   c. Reusable
   d. Effective and Efficient

   Why: _____

3. The Waterfall model of software development is

   a. A reasonable approach when requirements are well defined.
   b. A good approach when a working program is required quickly
   c. The best approach to use for projects with large development teams
   d. An old fashioned model that is rarely used any more.

   Why: _____

4. The Waterfall model of software development is also known as the

   a. Classical life cycle model
   b. Fountain model
   c. Spiral model
   d. Chaos model

   Why: _____

5. The prototyping model of software development is

   a. A reasonable approach when requirements are well defined
   b. A useful approach when a customer cannot define requirements clearly
   c. The best approach to use for projects with large development teams.
   d. A risky model that rarely produces a meaningful product

Why: _____

6. The rapid application development model is

   a. Another name for component-based development
   b. Useful approach when a customer cannot define requirements clearly
   c. A high speed adaptation of the linear sequential model
   d. All of the above

Why: _____

7. Evolutionary software process models

   a. Are iterative in nature
   b. Can easily accommodate product requirements changes
   c. Do not generally produce throw away systems
   d. All of the above

Why: _____

8. The spiral model of software development

   a. Ends with the delivery of the software product
   b. Includes project risks evaluation during each iteration
   c. All of the above

Why: _____

9. Which of the following is an Agile model:

   a. XP
   b. CMMI
   c. RAD

Why: _____

10. Which of the following is not considered a player in the software process?

   a. customers
   b. end-users
   c. project managers
   d. business development officers

Why: _____

SECTION B

11. Define and Give **EXAMPLE** of each of the following **(10 MARKS; 2 MARKS EACH)**

    I. Extreme Programming
    II. Refactoring
    III. Spike solution
    IV. Adaptive Software Development
    V. Component based Software development

12. Elaborate how a Spiral Model can be used as a Software Development Process Model. Identify its phases and the deliverables for each phase **(5 Marks)**

13. Describe SCRUM model, list down and describe each of its development actions **(5 Marks)**

14. You are required to design a Booking Software for "Kaloo Booking Agents". The requirements told are that it should be a web based application, allowing any customer to book a seat on any mode of transportation (Train, Bus or Aeroplane) from locations to and from Pakistan only on a predefined date and time as per the schedule. Tariff and Booking is generated ~~which is valid for only 6 hours,~~ 7 days prior to departure. The booking reference number is sent via SMS and the same reference number with original NIC is required to purchase the ticket by paying in cash at designated UBL Omni Stores. **(20 Marks)**

    payment is to be done within 6 hours of booking time ~~the departure~~ otherwise the booking will be cancelled.

    What are the Functional Requirements?

    What are Non-Functional Requirements?

    What are the Reliability Features of this system?

    What are Constraint Requirements?

---------- GOOD LUCK ----------